1	$h = 0.2$ $\frac{1}{2} \times "0.2" \times \left[9.2 + 8.6 + 2(8.4556 + 3.8512 + 5.0342 + 7.8297)\right]$ 6.814	B1 M1 A1
		(3 marks)

2a	$\left(f\left(\frac{3}{2}\right)=\right)\ 4\left(\frac{3}{2}\right)^3-8\left(\frac{3}{2}\right)^2+5\left(\frac{3}{2}\right)+a=0\Rightarrow a=...$	M1
	$\Rightarrow \frac{27}{2}-18+\frac{15}{2}+a=0\Rightarrow a=-3$ *	A1*
		(2)
b	Example where $2x-3$ is a linear factor: $2x-3\overline{)4x^3-8x^2+5x-3}$ $\quad 2x^2-x+1$ $4x^3-6x^2$ $-2x^2+5x$ $-2x^2+3x$ $+2x-3$ $+2x-3$ 0	M1A1
	$(-1)^2-4\times2\times1=-7<0\Rightarrow$ no real roots so $x=\frac{3}{2}$ is the only one real root *	dM1A1*
		(4)
		(6 marks)

3a(i)	$r = \sqrt{(8-2)^2 + (-3-5)^2} = 10$	M1A1
(ii)	$(x-2)^2 + (y-5)^2 = 100$	A1ft
		(3)
b	Gradient between centre and $P = -\frac{4}{3}$	B1
	Perpendicular gradient $= \frac{3}{4}$	M1
	$y+3 = \frac{3}{4}(x-8)$	M1
	$3x-4y-36 = 0$	A1
		(4)
		(7 marks)

4a	3^5 or 243	B1
		(1)
bi	$(B=)5\times3^4 p \ (=405p) \qquad (D=)10\times3^2 p^3 \ (=90p^3)$	M1A1
	$B=18D \Rightarrow =405p=18\times90p^3 \Rightarrow p^2=\frac{1}{4} \Rightarrow p=-\frac{1}{2}$	M1A1
ii	$(C=)\ {}^5C_2\times3^3\times\left(-\frac{1}{2}\right)^2 \ =\frac{135}{2}$	M1A1
		(6)
		(7 marks)

5	$\log_2 16x + \log_2(x+1) = 3 + \log_2(x+6)$	
	$\log_2 a = b \Rightarrow 2^b = a$	B1
	$\log_2 16x(x+1) = \log_2 8(x+6)$	M1
	$2x^2 + x - 6 = 0$	A1
	$(2x-3)(x+2) = 0$	dM1
	$x = \frac{3}{2}$ only	A1cso
		(5 marks)

6a	$\frac{k}{2k-15}=\frac{k+4}{k}\Rightarrow k^2=2k^2+8k-15k-60\Rightarrow k^2-7k-60=0$ *	M1A1*
		(2)
b	$k=12$	B1
	$\Rightarrow \frac{"12"}{2\times"12"-15}\ \left(=\frac{4}{3}\right)$	M1
	e.g. $("12"+4)\times\frac{4}{3}=...$ or $(2\times"12"-15)\times\left("\frac{4}{3}"\right)^3=...$	M1
	$=21300$	A1
		(4)
c	e.g. $\frac{9\left(1-\left(\frac{4}{3}\right)^N\right)}{1-\frac{4}{3}}>3000\Rightarrow\left(\frac{4}{3}\right)^N>\frac{1009}{9}$	M1
	$\log\left(\frac{4}{3}\right)^N>\log\left(\frac{1009}{9}\right)\Rightarrow N>\frac{\log\left(\frac{1009}{9}\right)}{\log\left(\frac{4}{3}\right)}$	dM1
	$N>16.405...\Rightarrow N=17$	A1
		(3)
		(9 marks)

7a	$H = t^{\frac{1}{2}} + \frac{3t^{\frac{3}{2}}}{10} - \frac{t^{\frac{5}{2}}}{20} + 17 \Rightarrow \left(\frac{\mathrm{d}H}{\mathrm{d}t} =\right) \frac{1}{2}t^{-\frac{1}{2}} + \frac{9}{20}t^{\frac{1}{2}} - \frac{1}{8}t^{\frac{3}{2}}$	M1A1A1
	e.g. $\frac{1}{2}t^{-\frac{1}{2}} + \frac{9}{20}t^{\frac{1}{2}} - \frac{1}{8}t^{\frac{3}{2}} = 0 \Rightarrow \frac{1}{2} + \frac{9}{20}t - \frac{1}{8}t^2 = 0 \ldots \Rightarrow 5\alpha^2 - 18\alpha - 20 = 0$ *	dM1A1*
		(5)
b	$\alpha = \frac{9+\sqrt{181}}{5} = \text{awrt } 4.491$	B1
		(1)
c	$\left(\frac{\mathrm{d}^2H}{\mathrm{d}t^2} =\right) -\frac{1}{4}t^{-\frac{3}{2}} + \frac{9}{40}t^{-\frac{1}{2}} - \frac{3}{16}t^{\frac{1}{2}} \Rightarrow -\frac{("4.49")^{-\frac{3}{2}}}{4} + \frac{9("4.49")^{-\frac{1}{2}}}{40} - \frac{3("4.49")^{\frac{1}{2}}}{16}$	M1
	$\frac{\mathrm{d}^2H}{\mathrm{d}t^2} = (-0.3\ldots) < 0 \Rightarrow \max$ *	A1*
		(2)
		(8 marks)

8(i)	Eg $7^2+11^2=170 \Rightarrow 170$ is a multiple of 10 so the statement is untrue *	M1A1*
		(2)
(ii)	Check first if they have multiplied the inequality e.g. by 4 to give $4<4x^2-xy<60$	M1A1
	Concludes that $1<x^2-\frac{xy}{4}<15$ for all x and y that are positive even integers less than 6 *	A1*
		(3)
		(5 marks)

x	y	$x^2-\frac{xy}{4}$ or $4x^2-xy$	
2	2	3	12
2	4	2	8
4	2	14	56
4	4	12	48

9a	$3\cos\theta(\tan\theta\sin\theta+3)=11-5\cos\theta$ $3\sin^2\theta+9\cos\theta=11-5\cos\theta$ or $3\cos\theta\left(\frac{\sin\theta}{\cos\theta}\sin\theta+3\right)=11-5\cos\theta$ $3(1-\cos^2\theta)+9\cos\theta=11-5\cos\theta$ $3\cos^2\theta-14\cos\theta+8=0$ *	M1 dM1 A1*
		(3)
b	$\cos 2x=\frac{2}{3}$ $x=\frac{\cos^{-1}\left(\frac{2}{3}\right)}{2}=\ldots$ $x=$ awrt 24.1, awrt 155.9, awrt 204.1, awrt 335.9	B1 M1 A1A1
		(4)
		(7 marks)

10a	$\dfrac{(x-k)^2}{\sqrt{x}} = \dfrac{x^2-2kx+k^2}{x^{\frac{1}{2}}} = \dfrac{x^2}{x^{\frac{1}{2}}} - \dfrac{2kx}{x^{\frac{1}{2}}} + \dfrac{k^2}{x^{\frac{1}{2}}} = x^{\frac{3}{2}} - 2kx^{\frac{1}{2}} + k^2x^{-\frac{1}{2}}$	M1
	$\int_1^{16} x^{\frac{3}{2}} - 2kx^{\frac{1}{2}} + k^2x^{-\frac{1}{2}}\ \mathrm{d}x = \left[\dfrac{2}{5}x^{\frac{5}{2}} - \dfrac{4}{3}kx^{\frac{3}{2}} + 2k^2x^{\frac{1}{2}}\right]_1^{16}$	M1A1
	$\dfrac{2}{5}(16)^{\frac{5}{2}} - \dfrac{4}{3}k(16)^{\frac{3}{2}} + 2k^2(16)^{\frac{1}{2}} - \dfrac{2}{5} + \dfrac{4}{3}k - 2k^2 = 6k^2 - 84k + \dfrac{2046}{5}$	dM1A1
		(5)
b	$9 = \dfrac{(1-k)^2}{\sqrt{1}} \Rightarrow 9 = (1-k)^2$	M1
	$k = 4$ only *	A1*
		(2)
c	$q = 36$	B1
	$\dfrac{1}{2}(9+"36")\times 15 - \left("6"(4)^2 - "84"(4) + \dfrac{2046}{5}\right) = \dfrac{1683}{10}$	M1A1
		(3)
		(10 marks)

11a	$u_2 = b - 3a$ $u_3 = b - a(b-3a) \quad \left(= b - ab + 3a^2\right)$	B1 B1
		(2)
b	$3 + b - 3a + b - a(b-3a) = 153$ oe $3 + (a+9) - 3a + (a+9) - a((a+9) - 3a) = 153 \Rightarrow 2a^2 - 10a - 132 = 0$ $\Rightarrow a^2 - 5a - 66 = 0$ *	M1 dM1A1*
		(3)
c	$a = -6$ $b = 9 - 6 = 3 \Rightarrow u_2 = 3 - 3(-6) = 21$ or $u_2 = 9 - 2a = 9 - 2 \times -6 = 21$	B1 M1A1
		(3)
		(8 marks)